J SCALING EFFECTS

Can we consistently locate function vectors given various sizes of a single language model architecture? We test this by observing all sizes of Llama 2, ranging from 7B parameters to 70B. We use the same methods as in §3.1, adding function vectors to each layer of the model and observing accuracy on our subset of 6 tasks at each layer.

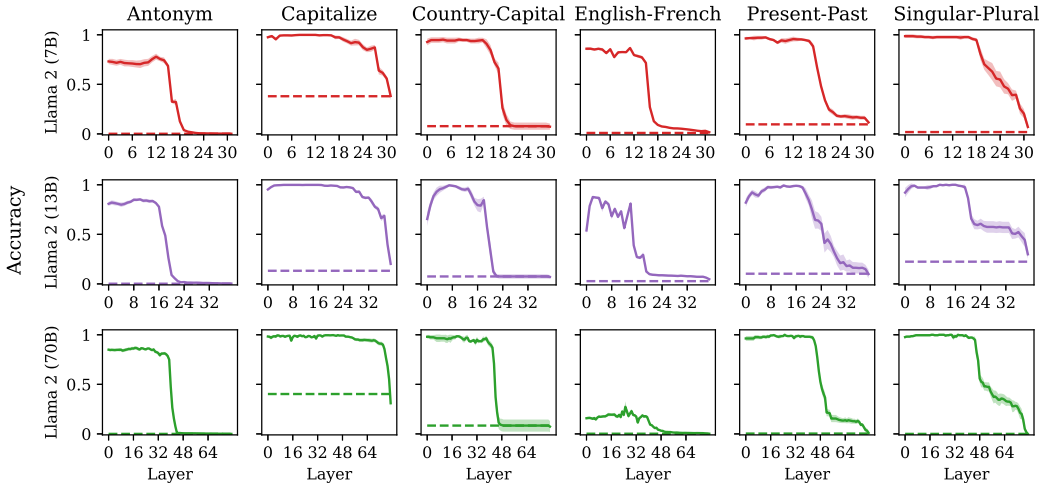

Figure 26: Zero-shot accuracy across Llama 2 model sizes in zero-shot settings. We show accuracies before adding the function vector (dotted lines) and after adding the FV to a specific layer (solid lines).

We find that results (Figure 26) are largely consistent across model sizes. Function vectors generally result in the highest zero-shot accuracies when added to the early to middle layers; this is true regardless of the total number of layers in the model.

K Composition on Other Models

In this section we include additional composition results for Llama 2 13B and 70B models in Tables 20 and 21 respectively.

Table 20: The accuracy of ICL, calculated FV v_{BD} zero-shot interventions, and vector-composed v^*_{BD} zero-shot interventions when performing several list-oriented tasks on LLaMA 2 (13B).

Task	ICL (ten-shot)	v_{BD} (FV on zero-shot)	v^*_{BD} (sum on zero-shot)
Last-Antonym	0.53 ± 0.02	0.26 ± 0.03	0.17 ± 0.02
Last-Capitalize	0.94 ± 0.01	0.63 ± 0.03	0.70 ± 0.03
Last-Country-Capital	0.86 ± 0.02	0.73 ± 0.02	0.37 ± 0.04
Last-English-French	0.75 ± 0.02	0.32 ± 0.02	0.12 ± 0.02
Last-Present-Past	0.96 ± 0.01	0.22 ± 0.02	0.24 ± 0.02
Last-Singular-Plural	0.89 ± 0.01	0.43 ± 0.03	0.53 ± 0.03
Last-Capitalize-First-Letter	0.85 ± 0.02	0.89 ± 0.02	0.89 ± 0.02
Last-Product-Company	0.47 ± 0.01	0.44 ± 0.03	0.60 ± 0.03

Table 21: LLaMA 2 (70B)

Task	ICL (ten-shot)	v_{BD} (FV on zero-shot)	v^*_{BD} (sum on zero-shot)
Last-Antonym	0.67 ± 0.03	0.43 ± 0.03	0.47 ± 0.03
Last-Capitalize	0.99 ± 0.00	0.93 ± 0.01	0.95 ± 0.01
Last-Country-Capital	0.81 ± 0.03	0.91 ± 0.02	0.94 ± 0.02
Last-English-French	0.84 ± 0.02	0.13 ± 0.01	0.17 ± 0.03
Last-Present-Past	0.98 ± 0.01	0.93 ± 0.01	0.94 ± 0.01
Last-Singular-Plural	0.98 ± 0.01	0.69 ± 0.04	0.69 ± 0.04
Last-Capitalize-First-Letter	0.67 ± 0.03	0.60 ± 0.02	0.68 ± 0.03
Last-Product-Company	0.49 ± 0.02	0.34 ± 0.01	0.34 ± 0.03

L EVALUATING FUNCTION VECTORS ON CYCLIC TASKS

In Appendix A, we discuss whether function vectors (FVs) can be thought of as simple word vector offsets, and show that cyclic tasks (such as antonyms) are a counterexample to this claim. In this section we report the causal effects of function vectors on two additional tasks with cyclic subsets —"next-item" and "previous-item" — providing further evidence that function vectors are not just simple semantic vector offsets but instead can be thought of as a trigger of nontrivial functions.

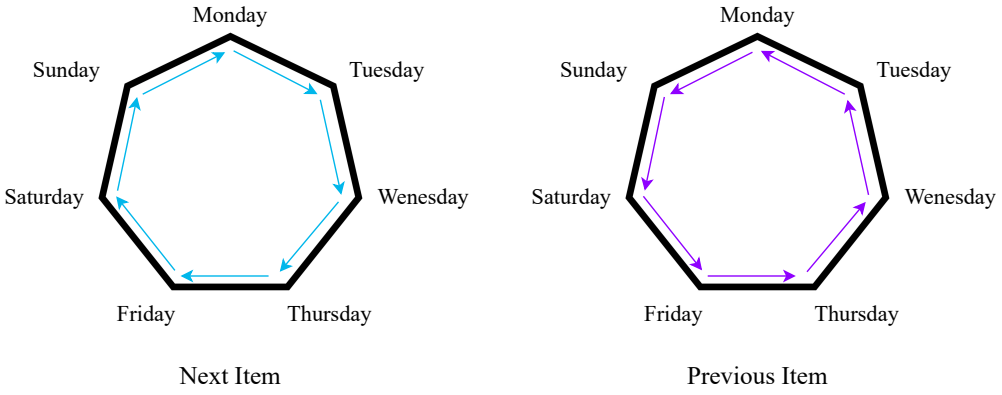

Figure 27: An example of cyclic structure for days of the week, which is a subset of the data for both the Next-Item and Previous-Item tasks. The cycles in each task follow the opposite order (e.g. Next-Item(`Monday`) = `Tuesday`, but Previous-Item(`Monday`) = `Sunday`.

The "next-item" task contains pairs of words which are related via the abstract idea of "next". The "previous-item" task contains the reciprocal version of the word pairs in the "next-item" task. Flipping the direction in this manner means each pair communicates the idea of "previous" instead. Both tasks are collected over a heterogeneous set of sequential data that includes cyclic types such as days of the week, months of the year, and letters of the alphabet, as well as non-cyclic types such as numbers and roman numerals. We include samples of example data pairs for these datasets in Appendix E.

However, a single ICL example for the Previous-Item task might look like "`Q: Friday\nA: Thursday\n\nQ: six\nA: five\n\nQ: a\nA:z\n\nQ: VII\nA:VI\n\nQ: September\nA:`". The model would ideally be able to answer "August" given this ICL prompt.

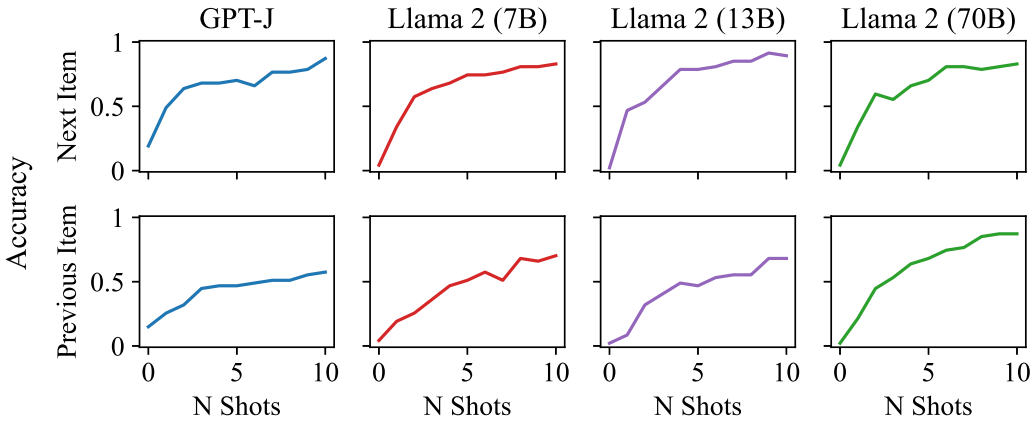

Figure 28: ICL performance on the "Next-Item" and "Previous-Item" cyclic tasks for 4 different models. The performance is usually better for the next-item task than for the previous-item task. However, 10-shot performance suggests these models are able to perform these tasks fairly well.